

domain ↗ ↘ codomain



Which of the following functions belong to $\mathcal{F}(\mathbb{R}, \mathbb{R})$?

- 1. $f_1(x) = x$ for all $x \in \mathbb{R}$
- 2. $f_2(x) = \sin x$ for all $x \in \mathbb{R}$
- 3. $f_3(x) = 1$ if $x = 0$ and $f_3(x) = 0$ if $x \neq 0$
- 4. $f_4(x) = i$ if $x = 0$ and $f_4(x) = 0$ if $x \neq 0$

Watch start of
lecture on
NUCLEAR

function as
"tuple"/"vector"

If $f \in \mathcal{F}(\mathbb{R}, \mathbb{R})$, then all entries in the row of f
should be a real number

Domain x	-2	$-\frac{\pi}{2}$	-1	0	1	$\frac{\pi}{2}$	2		
f_1	-2	$-\frac{\pi}{2}$	-1	0	1	$\frac{\pi}{2}$	2		
f_2	$\sin(-2)$	-1	$\sin(-1)$	0	$\sin(1)$	1	$\sin 2$		
f_3	0	0	1	1	0	0	0		
f_4	0	0	0	i	0	0	0		
$f_1 \oplus f_3$	-2	$-\frac{\pi}{2}$	-1	1	1	$\frac{\pi}{2}$	2		

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Lecture 01B

What can be a Vector?

MH1201 Linear Algebra 2

<https://www.wooclap.com/>

HIDDEN

Lecture 01 – Vector Spaces

- Vector Operations
- Examples of Vector Space
 - $\mathcal{M}_{m \times n}(\mathbb{F})$: $m \times n$ -Matrices
 - $\mathcal{F}(X, \mathbb{F})$: Functions from X to $\mathbb{F}$
- Axioms of Vector Space
- Properties of Vector Space
- Subspace

Axioms of Vector Space

Axioms for Vector Space

Vector Space. $(\mathcal{V}, \oplus, \otimes)$ is a vector space if all the following hold.

- **(Closed under $\oplus$).** For all $u, v \in \mathcal{V}$, we have $u \oplus v \in \mathcal{V}$.
- **(Closed under $\otimes$).** For all $\lambda \in \mathbb{F}, u \in \mathcal{V}$, we have $\lambda \otimes u \in \mathcal{V}$.

Vector Addition $\oplus$

- **($\oplus$ is commutative).** For all $u, v \in \mathcal{V}$,
$$u \oplus v = v \oplus u.$$
- **($\oplus$ is associative).** For all $u, v, w \in \mathcal{V}$,
$$(u \oplus v) \oplus w = u \oplus (v \oplus w).$$
- **(Additive Identity).** There exists $0 \in \mathcal{V}$ such that
$$u \oplus 0 = 0 \oplus u = u \text{ for all } u \in \mathcal{V}.$$
- **(Additive Inverse).** For all $u \in \mathcal{V}$, there exists $u^* \in \mathcal{V}$ such that
$$u \oplus u^* = u^* \oplus u = 0.$$

Scalar Multiplication $\otimes$

- **($\otimes$ is associative).** For all $\lambda, \mu \in \mathbb{F}, u \in \mathcal{V}$,
$$\lambda \otimes (\mu \otimes u) = (\lambda\mu) \otimes u.$$
- **(Multiplicative Identity).** Consider $1 \in \mathbb{F}$. Then
$$1 \otimes u = u \text{ for all } u \in \mathcal{V}.$$
- **(Distributive-1).** For all $\lambda \in \mathbb{F}, u, v \in \mathcal{V}$,
$$\lambda \otimes (u \oplus v) = (\lambda \otimes u) \oplus (\lambda \otimes v).$$
- **(Distributive-2).** For all $\lambda, \mu \in \mathbb{F}, u \in \mathcal{V}$,
$$(\lambda + \mu) \otimes u = (\lambda \otimes u) \oplus (\mu \otimes u).$$

Important: Vector space has three things. $\left. \begin{array}{l} \mathcal{V}, \oplus, \otimes \\ \swarrow \\ \text{the space itself} \end{array} \right\} \text{define } \oplus, \otimes \text{ s.t. all axioms are obeyed.}$

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

Vector Space. $(\mathcal{V}, \oplus, \otimes)$ is a vector space if all the following hold...

- (Closed under $\oplus$). For all $u, v \in \mathcal{V}$,

$$u \oplus v \in \mathcal{V}.$$

- (Closed under $\otimes$). For all $\lambda \in \mathbb{F}, u \in \mathcal{V}$,

$$\lambda \otimes u \in \mathcal{V}.$$

(8 other axioms hidden)

- $\mathcal{M}_{m \times n}(\mathbb{F})$ is the set of $m \times n$ matrices whose entries belong to $\mathbb{F}$.

- $A \oplus B$ is defined such that

$$[a_{ij}] \oplus [b_{ij}] \triangleq [a_{ij} + b_{ij}]$$

- $\lambda \otimes A$ is defined such that

$$\lambda \otimes [a_{ij}] \triangleq [\lambda a_{ij}]$$

is equal by definition to

On Monday, we showed that $\mathcal{M}_{m \times n}$ is closed under $\oplus$ and $\otimes$.

- The rest of axioms: D.I.Y $\rightarrow$ very detailed solⁿ are on colab

Proposition. $(\mathcal{M}_{m \times n}(\mathbb{F}), \oplus, \otimes)$ is a vector space.

Vector Space. $(\mathcal{V}, \oplus, \otimes)$ is a vector space if all the following hold...

- **(Closed under $\oplus$).** For all $u, v \in \mathcal{V}$,

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$$\lambda \otimes u \in \mathcal{V}.$$

(8 other axioms hidden)

- $\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

- $(f \oplus g)(x)$ is defined such that

$$(f \oplus g)(x) \triangleq f(x) + g(x)$$

- $(\lambda \otimes f)(x)$ is defined such that

$$(\lambda \otimes f)(x) \triangleq \lambda f(x)$$

On Mon, we show that $\mathcal{F}(X, \mathbb{F})$ is closed under $\oplus$ and $\otimes$

- Today: we check some axioms

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

Vector Addition $\oplus$

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 $(u \oplus v) \oplus w = u \oplus (v \oplus w)$.
- (**Additive Identity**). There exists $0 \in \mathcal{V}$ such that
 $u \oplus 0 = 0 \oplus u = u$ for all $u \in \mathcal{V}$.
- (**Additive Inverse**). For all $u \in \mathcal{V}$, there exists $u^* \in \mathcal{V}$ such that
 $u \oplus u^* = u^* \oplus u = 0$.

When are the two functions equal?
 $f = g$ iff $f(x) = g(x)$ for all $x \in X$
e.g. $f(x) = \sin^2 x + \cos^2 x$
 $g(x) = 1$

WTS: $f \oplus g = g \oplus f$ for all $f, g \in \mathcal{F}(X, \mathbb{F})$
i.e. $(f \oplus g)(x) = (g \oplus f)(x)$ for all x
LHS = $(f \oplus g)(x) = f(x) + g(x)$
RHS = $(g \oplus f)(x) = g(x) + f(x)$ → property of numbers
Since $f(x) + g(x) = g(x) + f(x)$ for all x ,
LHS = RHS for all x
 $\therefore f \oplus g = g \oplus f$

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

Vector Addition $\oplus$

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 $u \oplus u^* = u^* \oplus u = 0$.

Subtle point:

$$(f \oplus g)(x) = f(x) + g(x)$$

↓
vector
addition

↓
addition of
numbers

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in $\mathbb{F}$ to elements in $\mathbb{F}$.

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 $u \oplus u^* = u^* \oplus u = 0$.

(Additive identity) For $f \in \mathcal{F}(X, \mathbb{F})$,
there exists $\hat{0} \in \mathcal{F}(X, \mathbb{F})$
st $f \oplus \hat{0} = \hat{0} \oplus f = f$

Given f ,
WTF $z \in \mathcal{F}$ st
 $(z \oplus f)(x) = f(x)$
 $(f \oplus z)(x) = f(x)$ for all x

$\rightarrow$ Reverse Engineer
Thought Process:
LHS = $(z \oplus f)(x)$
 $= z(x) + f(x)$
RHS = $f(x)$
 $\therefore z(x)$ should be 0

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

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Additive Identity

For all $f \in \mathcal{F}$,

choose $z(x)$ to be the function $z(x) = 0$ for all x .

Check: For all x , LHS = $(f \oplus z)(x)$

$$= f(x) + z(x) = f(x) + 0$$

$$= f(x) = \text{RHS}$$

$$\therefore f \oplus z = f.$$

P1: Check $z \oplus f = f$

Actual Proof

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in $\mathbb{F}$ to elements in $\mathbb{F}$.

Vector Addition $\oplus$

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 $u \oplus u^* = u^* \oplus u = 0$.

$\downarrow$
 $z(x)$

(Additive inverse) For all $f \in \mathcal{F}$,
WTF $f^*(x)$

$$\left. \begin{aligned} f \oplus f^* &= z(x) \\ f^* \oplus f &= z(x) \end{aligned} \right\} \text{ for all } x$$

What do we know about $f^*(x)$?

$$\begin{aligned} \text{LHS} &= (f \oplus f^*)(x) = f(x) + f^*(x) \\ \text{RHS} &= z(x) = 0 \end{aligned} \left. \right\} \Rightarrow f^*(x) = -f(x)$$

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in $\mathbb{F}$ to elements in $\mathbb{F}$.

Vector Addition $\oplus$

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 $u \oplus 0 = 0 \oplus u = u$ for all $u \in \mathcal{V}$.
- (Additive Inverse). For all $u \in \mathcal{V}$, there exists $u^* \in \mathcal{V}$ such that
 $u \oplus u^* = u^* \oplus u = 0$.

(Additive Inverse) For all $f \in \mathcal{F}$,
choose $f^*(x) = -f(x)$ for all x .
check that $LHS = (f \oplus f^*)(x)$
 $= f(x) + f^*(x)$
 $= f(x) - f(x)$
 $= 0 = z(x)$
 $= RHS$

$$\forall \gamma: f^* \oplus f = z$$

$$\text{for all } x. \quad \therefore f \oplus f^* = z$$

Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

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Proposition. $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$ is a vector space.

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in $\mathbb{F}$ to elements in $\mathbb{F}$.

Scalar Multiplication $\otimes$

- **($\otimes$ is associative).** For all $\lambda, \mu \in \mathbb{F}, \mathbf{u} \in \mathcal{V}$,
$$\lambda \otimes (\mu \otimes \mathbf{u}) = (\lambda\mu) \otimes \mathbf{u}.$$
- **(Multiplicative Identity).** Consider $1 \in \mathbb{F}$. Then
$$1 \otimes \mathbf{u} = \mathbf{u} \text{ for all } \mathbf{u} \in \mathcal{V}.$$
- **(Distributive-1).** For all $\lambda \in \mathbb{F}, \mathbf{u}, \mathbf{v} \in \mathcal{V}$,
$$\lambda \otimes (\mathbf{u} \oplus \mathbf{v}) = (\lambda \otimes \mathbf{u}) \oplus (\lambda \otimes \mathbf{v}).$$
- **(Distributive-2).** For all $\lambda, \mu \in \mathbb{F}, \mathbf{u} \in \mathcal{V}$,
$$(\lambda + \mu) \otimes \mathbf{u} = (\lambda \otimes \mathbf{u}) \oplus (\mu \otimes \mathbf{u}).$$

Four more Axioms: DIY

Additive Identity and Additive Inverse

(Additive Identity). There exists $\mathbf{0} \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{0} = \mathbf{0} \oplus \mathbf{u} = \mathbf{u} \text{ for all } \mathbf{u} \in \mathcal{V}.$$

$\mathbf{0}$ is also called the zero vector.

(Additive Inverse). For all $\mathbf{u} \in \mathcal{V}$, there exists $\mathbf{u}^* \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{u}^* = \mathbf{u}^* \oplus \mathbf{u} = \mathbf{0}.$$

$\mathbf{u}^*$ is the negative vector of $\mathbf{u}$.

$\rightarrow$ functions
 $F(X, \mathbb{R})$

• $\mathbf{z}$: Zero function

• negative of f is f^*
 $f^*(x) = -f(x)$

$\rightarrow$ matrices
 $M_{m \times n}(\mathbb{R})$

• $\hat{\mathbf{0}}_{m \times n} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \vdots & \vdots & \vdots \end{bmatrix}$
 $\hookrightarrow$ zero matrix

• negative of $\hat{A} = [a_{ij}]$ is
 $[-a_{ij}]$

Properties of Vector Space

Why Not More Axioms?

Why not include the following “obvious axioms”?

- $0 \otimes \mathbf{u} = \mathbf{0}$ for all $\mathbf{u} \in \mathcal{V}$.
- $\lambda \otimes \mathbf{0} = \mathbf{0}$ for all $\lambda \in \mathbb{F}$.
- $\lambda \otimes \mathbf{v} = \mathbf{0}$ if and only if $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$.
- $-1 \otimes \mathbf{v}$ is the negative vector of $\mathbf{v}$.

these are typical arithmetic properties that we always use.

Ans: these properties can be derived/proved using the axioms.

Some Properties over Vector Spaces

- The zero vector is unique. *will go thru*
- For any vector, its negative vector is unique. *DIY*
- $0 \otimes \mathbf{u} = \mathbf{0}$ for all $\mathbf{u} \in \mathcal{V}$. *will go thru*
- $\lambda \otimes \mathbf{0} = \mathbf{0}$ for all $\lambda \in \mathbb{F}$. *DIY*
- $\lambda \otimes \mathbf{v} = \mathbf{0}$ if and only if $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$. *DIY*
- $-1 \otimes \mathbf{v}$ is the negative vector of $\mathbf{v}$. *DIY*

** DIY stuff have solutions on colab*

Proposition. The zero vector is unique.

* There is only one zero vector

Suppose $\underline{0}$ and $\underline{0}'$ are zero vectors in $\mathcal{V}$.

WTS $\underline{0} = \underline{0}'$

$$\underline{0} = \underline{0} \oplus \underline{0}' = \underline{0}'$$

Because $\underline{0}'$ is a zero vector

$\because \underline{0}$ is a zero vector

$\therefore \underline{0} \& \underline{0}'$ are the same vector

(Additive Identity). There exists $\mathbf{0} \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{0} = \mathbf{0} \oplus \mathbf{u} = \mathbf{u} \text{ for all } \mathbf{u} \in \mathcal{V}.$$

$\mathbf{0}$ is also called the **zero vector**.

(Additive Inverse). For all $\mathbf{u} \in \mathcal{V}$, there exists $\mathbf{u}^* \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{u}^* = \mathbf{u}^* \oplus \mathbf{u} = \mathbf{0}.$$

$\mathbf{u}^*$ is the negative vector of $\mathbf{u}$.

Proposition. For any vector, its negative vector is unique.

Let u be a vector.

Let v and v' be negative vectors of u

$$v = v \oplus 0 = v \oplus (u \oplus v') = (v \oplus u) \oplus v' = 0 \oplus v' = v'$$

Therefore v and v' are the same.

(Additive Identity). There exists $0 \in \mathcal{V}$ such that

$$u \oplus 0 = 0 \oplus u = u \text{ for all } u \in \mathcal{V}.$$

0 is also called the **zero vector**.

(Additive Inverse). For all $u \in \mathcal{V}$, there exists $u^* \in \mathcal{V}$ such that

$$u \oplus u^* = u^* \oplus u = 0.$$

u^* is the negative vector of u .

Proposition. $0 \otimes u = 0$ for all $u \in V$.

$$\begin{aligned}
 0 \otimes u &= (0+0) \otimes u && \text{because } 0 = 0+0 \text{ adding numbers} \\
 &= (0 \otimes u) \oplus (0 \otimes u) && \text{because of distributive 2} \\
 &&& \text{adding vectors}
 \end{aligned}$$

Let x be the negative of $(0 \otimes u)$ & add x to both sides.

$$\text{LHS} = (0 \otimes u) \oplus u = 0$$

$$\begin{aligned}
 \text{RHS} &= (0 \otimes u) \oplus ((0 \otimes u) \oplus u) = (0 \otimes u) \oplus 0 \\
 &= 0 \otimes u
 \end{aligned}$$

doesn't matter where we put bracket cos addition is associative

(Additive Identity). There exists $0 \in V$ such that

$$u \oplus 0 = 0 \oplus u = u \text{ for all } u \in V.$$

0 is also called the **zero vector**.

(Additive Inverse). For all $u \in V$, there exists $u^* \in V$ such that

$$u \oplus u^* = u^* \oplus u = 0.$$

u^* is the negative vector of u .

(Distributive-1). For all $\lambda \in \mathbb{F}, u, v \in V$,

$$\lambda \otimes (u \oplus v) = (\lambda \otimes u) \oplus (\lambda \otimes v).$$

(Distributive-2). For all $\lambda, \mu \in \mathbb{F}, u \in V$,

$$(\lambda + \mu) \otimes u = (\lambda \otimes u) \oplus (\mu \otimes u).$$

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$$\therefore 0 \otimes u = 0$$

got ✓ correct line

Proposition. $\lambda \otimes \mathbf{0} = \mathbf{0}$ for all $\lambda \in \mathbb{F}$.

$$\begin{aligned}\lambda \otimes \underline{0} &= \lambda \otimes (\underline{0} + \underline{0}) \\ &= (\lambda \otimes \underline{0}) \oplus (\lambda \otimes \underline{0})\end{aligned}$$

Let χ be the negative of $\lambda \otimes \underline{0}$ & add χ on both sides.

$$\text{LHS} = (\lambda \otimes \underline{0}) + \chi = \underline{0}$$

$$\text{RHS} = (\chi \oplus (\lambda \otimes \underline{0})) \oplus (\lambda \otimes \underline{0}) = \underline{0} \oplus (\lambda \otimes \underline{0}) = \lambda \otimes \underline{0}$$

$$\therefore \lambda \otimes \underline{0} = \underline{0}$$

(Additive Identity). There exists $\mathbf{0} \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{0} = \mathbf{0} \oplus \mathbf{u} = \mathbf{u} \text{ for all } \mathbf{u} \in \mathcal{V}.$$

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(Additive Inverse). For all $\mathbf{u} \in \mathcal{V}$, there exists $\mathbf{u}^* \in \mathcal{V}$ such that

$$\mathbf{u} \oplus \mathbf{u}^* = \mathbf{u}^* \oplus \mathbf{u} = \mathbf{0}.$$

$\mathbf{u}^*$ is the negative vector of $\mathbf{u}$.

(Distributive-1). For all $\lambda \in \mathbb{F}, \mathbf{u}, \mathbf{v} \in \mathcal{V}$,

$$\lambda \otimes (\mathbf{u} \oplus \mathbf{v}) = (\lambda \otimes \mathbf{u}) \oplus (\lambda \otimes \mathbf{v}).$$

(Distributive-2). For all $\lambda, \mu \in \mathbb{F}, \mathbf{u} \in \mathcal{V}$,

$$(\lambda + \mu) \otimes \mathbf{u} = (\lambda \otimes \mathbf{u}) \oplus (\mu \otimes \mathbf{u}).$$

→ two way proof!

Proposition. $\lambda \otimes v = 0$ if and only if $\lambda = 0$ or $v = 0$.

If $\lambda = 0$, then $\lambda \otimes v = 0$ (Just proved)

If $v = 0$, then $\lambda \otimes v = 0$ (Just proved)

If $\lambda \neq 0$, $v \neq 0$, we prove by contradiction that $\lambda \otimes v \neq 0$

Suppose otherwise $\lambda \otimes v = 0$ ← assumption is wrong!

Because $\lambda \neq 0$, $\frac{1}{\lambda}$ exists. Multiply both sides by $\frac{1}{\lambda}$.

$$\text{LHS} = \frac{1}{\lambda} \otimes (\lambda \otimes v) = (\frac{1}{\lambda} \lambda) \otimes v = 1 \otimes v = v \quad \left. \vphantom{\frac{1}{\lambda} \otimes (\lambda \otimes v)} \right\} v = 0$$

$$\text{RHS} = \frac{1}{\lambda} \otimes 0 = 0$$

*contradiction

"If $\lambda \otimes v = 0$, then $\lambda = 0$ or $v = 0$ "

Contrapositive:

"If $\lambda \neq 0$ and $v \neq 0$, then $\lambda \otimes v \neq 0$ "

Proposition. $-1 \otimes v$ is the negative vector of v .

Proof

$$(-1 \otimes v) \oplus v$$

$$= (-1 \otimes v) \oplus (1 \otimes v)$$

$$= (-1 + 1) \otimes v$$

$$= 0 \otimes v$$

$$= 0$$

$\therefore -1 \oplus v$ is ~~a~~ ^{the} negative vector of v

because negative vectors are unique!

$$(\hat{A})^* = -1 \otimes \hat{A}$$

$$f^* = -1 \otimes f$$

Ten Axioms Again?

Vector Subspace

Vector Subspace.

Let $(\mathcal{V}, \oplus, \otimes)$ be a vector space.

Let $\mathcal{U}$ be a subset of $\mathcal{V}$.

We say $\mathcal{U}$ is a **subspace** if $(\mathcal{U}, \oplus, \otimes)$ is a vector space.

Question: How do we show that $\mathcal{U}$ is a subspace?

- Naive way: check all ten axioms
→ but this is not necessary because $\mathcal{V}$ is a vector space.

✓ change to $\mathcal{U}$.

Checking for a subspace??

Vector Space. $(\mathcal{U}, \oplus, \otimes)$ is a vector space if all the following hold.

- **(Closed under $\oplus$).** For all $u, v \in \mathcal{U}$, we have $u \oplus v \in \mathcal{U}$.
- **(Closed under $\otimes$).** For all $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have $\lambda \otimes u \in \mathcal{U}$.

Vector Addition $\oplus$

- **($\oplus$ is commutative).** For all $u, v \in \mathcal{U}$,
$$u \oplus v = v \oplus u.$$
- **($\oplus$ is associative).** For all $u, v, w \in \mathcal{U}$,
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- **(Additive Identity).** There exists $0 \in \mathcal{U}$ such that
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Scalar Multiplication $\otimes$

- **($\otimes$ is associative).** For all $\lambda, \mu \in \mathbb{F}$, $u \in \mathcal{U}$,
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- **(Distributive-1).** For all $\lambda \in \mathbb{F}$, $u, v \in \mathcal{U}$,
$$\lambda \otimes (u \oplus v) = (\lambda \otimes u) \oplus (\lambda \otimes v).$$
- **(Distributive-2).** For all $\lambda, \mu \in \mathbb{F}$, $u \in \mathcal{U}$,
$$(\lambda + \mu) \otimes u = (\lambda \otimes u) \oplus (\mu \otimes u).$$

Checking for a subspace??

Remember: $u \in \mathcal{U} \Rightarrow u \in \mathcal{V}$

$$u, v \in \mathcal{U} \Rightarrow u, v \in \mathcal{V}$$

$$\Rightarrow u + v = v + u$$

Vector Addition $\oplus$

- ($\oplus$ is commutative). For all $u, v \in \mathcal{U}$,
 $u \oplus v = v \oplus u \rightarrow \because u, v \in \mathcal{V}$

- ($\oplus$ is associative). For all $u, v, w \in \mathcal{U}$,
 $(u \oplus v) \oplus w = u \oplus (v \oplus w) \rightarrow \because u, v, w \in \mathcal{V}$

- (Additive Identity). There exists $0 \in \mathcal{U}$
such that

$$u \oplus 0 = 0 \oplus u = u \text{ for all } u \in \mathcal{U}.$$

- (Additive Inverse). For all $u \in \mathcal{U}$, there
exists $u^* \in \mathcal{U}$ such that

$$u \oplus u^* = u^* \oplus u = 0.$$

outstanding items

$\Rightarrow$ additive identity

$\&$ additive inverse

Checking for a subspace??

$u \in \mathcal{U}$ implies that $u \in \mathcal{V}$

Scalar Multiplication $\otimes$

- ($\otimes$ is associative). For all $\lambda, \mu \in \mathbb{F}$, $u \in \mathcal{U}$,
$$\lambda \otimes (\mu \otimes u) = (\lambda\mu) \otimes u.$$
- (Multiplicative Identity). Consider $1 \in \mathbb{F}$. Then
$$1 \otimes u = u \text{ for all } u \in \mathcal{U}.$$
- (Distributive-1). For all $\lambda \in \mathbb{F}$, $u, v \in \mathcal{U}$,
$$\lambda \otimes (u \oplus v) = (\lambda \otimes u) \oplus (\lambda \otimes v).$$
- (Distributive-2). For all $\lambda, \mu \in \mathbb{F}$, $u \in \mathcal{U}$,
$$(\lambda + \mu) \otimes u = (\lambda \otimes u) \oplus (\mu \otimes u).$$

• All these axioms will hold for $\mathcal{U}$.

Checking for a subspace??

$u \in \mathcal{U}$ implies that $u \in \mathcal{V}$

Vector Space. $(\mathcal{U}, \oplus, \otimes)$ is a vector space if all the following hold.

- **(Closed under $\oplus$).** For all $u, v \in \mathcal{U}$, we have

$$u \oplus v \in \mathcal{U}. \rightarrow \text{need to check}$$

- **(Closed under $\otimes$).** For all $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have

$$\lambda \otimes u \in \mathcal{U}. \rightarrow \text{need to check}$$

$$u, v \in \mathcal{U} \Rightarrow u, v \in \mathcal{V} \Rightarrow u \oplus v \in \mathcal{V} \Rightarrow u \oplus v \in \mathcal{U}$$

$\downarrow$
not always true, so we need to check

Checking for a subspace??

$$u \in \mathcal{U} \text{ implies that } u \in \mathcal{V}$$

Vector Space. $(\mathcal{U}, \oplus, \otimes)$ is a vector space if all the following hold.

- (Closed under $\oplus$). For all $u, v \in \mathcal{U}$, we have

$$u \oplus v \in \mathcal{U}.$$

- (Closed under $\otimes$). For all $\lambda \in \mathbb{F}$, $u \in \mathcal{U}$, we have

$$\lambda \otimes u \in \mathcal{U}.$$

- (Additive Identity). There exists $0 \in \mathcal{U}$.

- (Additive Inverse). For all $u \in \mathcal{U}$, there exists $u^* \in \mathcal{U}$ such that

$$u \oplus u^* = u^* \oplus u = 0.$$

not necessary to check

negative of u is $(-1) \otimes u$.

Proposition : If $\mathcal{U}$ is closed under $\otimes$ then negative u always belongs to $\mathcal{U}$.
*make use of the last property on slide for

proof : Given u ,

Wts : negative u belongs to $\mathcal{U}$.

i.e. $(-1) \otimes u$ belongs to $\mathcal{U}$.

Because $\mathcal{U}$ is closed under $\otimes$.

Summary.

Conditions for Subspace

Theorem. (Conditions for Subspace) Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** $\mathbf{0}$ belongs to $\mathcal{U}$.
- **(Closure under Vector Addition).** For all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, we have $\mathbf{u} \oplus \mathbf{v} \in \mathcal{U}$.
- **(Closure under Scalar Multiplication).** For $\lambda \in \mathbb{F}$, $\mathbf{u} \in \mathcal{U}$, we have $\lambda \otimes \mathbf{u} \in \mathcal{U}$.

Examples of Subspaces

set of symmetric matrices

Proposition. $\mathcal{S}_{n \times n}(\mathbb{F})$ is a subspace of $(\mathcal{M}_{n \times n}(\mathbb{F}), \oplus, \otimes)$.

set of poly of deg at most d

Proposition. $\mathcal{P}_d(\mathbb{F})$ is a subspace of $(\mathcal{F}(\mathbb{F}, \mathbb{F}), \oplus, \otimes)$.

Proposition. The set of solutions of a homogeneous system is a subspace of $(\mathbb{F}^n, \oplus, \otimes)$.